Lethe (Echo portion)

What test–retest repeatability validates about conformal interval *scale* in IVIM — and what it cannot

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Provisional build. The *method* and its synthetic self-test (§3–4) are self-contained and machine-checked (SOLID). The *real-data* result (§6) consumes **Caliper**'s conformal ruler and is contextualised by **Fashion**'s calibrated posterior, **Gauge**'s repeatability baseline, and **Minos**'s decision lens — all in review. Every assumption-dependent number is marked [P] and is regenerated from seeded results by `consistency.py`. This is a **Lethe** result: an honest limitation, reported faithfully, not a failure.

Abstract

A conformal prediction interval can be validated for marginal coverage only where a ground-truth label exists. In quantitative MRI it usually does not. The one label-free signal that is genuinely available is *repeatability*: same-session scan–rescan acquisitions. We ask a precise, and deliberately narrow, question of a deployed conformal intravoxel-incoherent-motion (IVIM) interval: is it the right *size* to capture a measurement's own irreproducibility? We answer it with *test–retest interval coverage* — whether one scan's parameter estimate falls inside the other scan's deployed interval — reported with bias-corrected-and-accelerated (BCa) bootstrap confidence intervals. The statistic is *precision, not accuracy*: the test–retest difference cancels any bias common to both scans, so it certifies that an interval is scaled to measurement noise and is provably blind to systematic error. It is also distinct from, and orthogonal to, the published width-*rank*-tracks-repeatability check [P] (Spearman $r = 0.60$): rescaling every width by a constant leaves the rank correlation invariant but moves coverage arbitrarily. On a synthetic harness the statistic recovers its analytic behaviour exactly. On 76 real same-day test–retest breast-DWI tumors (ACRIN-6698), the conformally deployed interval for the well-identified diffusion coefficient D is about four times too *narrow* to contain real scan–rescan variation: coverage 0.263 (BCa 95% $[0.158, 0.355]$) against an analytic target of 0.755, scale ratio $R = 0.247$, under-scaled across the entire plausible signal-to-noise range. We conclude — as a constrained validation — that conventionally calibrated conformal IVIM intervals *order* repeatability correctly but are not *sized* to it, because region-level repeatability is dominated by non-thermal sources that synthetic calibration does not model.

1 Introduction

Distribution-free conformal prediction gives finite-sample marginal coverage *when a held-out label is available*. For quantitative MRI parameter maps — here the IVIM triplet (D, D^*, f) — no per-voxel ground truth exists in vivo, so coverage is, strictly, unmeasurable on real tissue. What *is* available, on a minority of protocols, is *repeatability*: a same-session "coffee-break" rescan that re-measures the same tissue. Repeatability is the workhorse of the Quantitative Imaging Biomarkers Alliance (QIBA) Profile framework, where the repeatability coefficient bounds the change in a biomarker that exceeds measurement noise.

Repeatability is *precision*, not accuracy or coverage. The central discipline of this note is to take that distinction literally and to extract from repeatability *only* what it can license. We define a single ground-truth-free statistic on a deployed conformal interval — whether it is correctly *sized* to scan–rescan variation — prove that it is invariant to bias by construction, prove that it is mathematically independent of the existing width-*ordering* check, validate it on controlled synthetic data, and then apply it to real test–retest IVIM. We additionally exhibit, on synthetic ground truth, a *constructive counterexample* (§5, SOLID) in which a region-level estimate is excellently repeatable yet its interval covers the *true* parameter far below nominal — turning the precision $\neq$ coverage limitation from asserted into shown. The verdict is negative and sharply scoped, which is why this work lives in **Lethe**, the project's home for honest-limitation results.

2 Background and positioning

Three companion efforts frame this note; all are in review, so every claim that leans on them is marked [P]. **Gauge**[P] brought split-conformal and conformalized quantile regression to IVIM and reported, on the same ACRIN-6698 test–retest arm used here, that the conformal interval *width* rank-tracks the scan–rescan variability of D (Spearman $r = 0.60$, 95% CI $[0.42, 0.72]$; for D^* the association is null, $r = -0.17$), explicitly framing it as "repeatability-tracking, not validation, accuracy, calibration, or coverage." **Fashion**[P] (retooled, in review at *NMR in Biomedicine*) established which uncertainty paradigms actually cover D^* and supplies the calibrated posterior a conformal width is built on; in its retooled form it leads with an assumption-free boundary-railing diagnostic and openly scopes its calibration ruler to a *secondary, ground-truth-only* instrument that under-covers D^* *conditionally* in the high-D^* regime — a bounded, conditional admission that reinforces the constrained-validation reading of this note: even the ruler's own paper now agrees that a ground-truth-free calibration certificate is scoped, not absolute. **Minos**[P] prices the *decision* value of a calibrated error bar, motivating why a correctly *sized* interval matters. We reuse **Caliper** (MIT, in-tree) for the conformal ruler, read-only.

Our contribution is the axis Gauge declined: not whether width *orders* repeatability, but whether it is the right *size* for it. Rank and scale are independent (§3); a width can order repeatability perfectly and still be the wrong magnitude.

3 Method

Setup. For tumor i we obtain two same-session repeat estimates $\hat{\theta}_i^A, \hat{\theta}_i^B$ of each IVIM parameter $\theta \in \{D, D^*, f\}$ from a segmented plug-in fit on the sparse four-b scheme $\{0, 100, 600, 800\}$ s/mm², and a deployed conformal interval $[\ell_i, u_i]$ with half-width w_i. All repeats are region-level (whole-tumor ROI mean); the two acquisitions are unregistered, so no per-voxel statement is licensed.

The statistic. The headline is the symmetrized *test–retest interval coverage*

$$c \;=\; \tfrac{1}{2n}\sum_{i=1}^{n}\Big(\mathbf{1}[\hat{\theta}_i^B \in [\ell_i^A, u_i^A]] + \mathbf{1}[\hat{\theta}_i^A \in [\ell_i^B, u_i^B]]\Big),$$

reported with a BCa bootstrap 95% CI over tumors. We also report the scale ratio $R = \sigma_{\text{impl}}/\sigma_{\text{rep}}$, where $\sigma_{\text{impl}} = w/2z_{1-\alpha/2}$ is the measurement scale implied by the interval and σ_{rep} is the robust repeatability standard deviation estimated from the test–retest differences.

Precision, not accuracy (the legitimacy razor). Write each estimate as $\hat{\theta} = \theta_{\text{true}} + b + \varepsilon$, with b a systematic bias and ε measurement noise. The test–retest difference $\Delta = \hat{\theta}^B - \hat{\theta}^A = \varepsilon^B - \varepsilon^A$ *cancels b exactly*. Every statistic here therefore certifies only that an interval is sized to measurement irreproducibility (precision); it is provably blind to accuracy and makes no ground-truth coverage claim. A consequence we use as a sanity anchor: a correctly *measurement-scaled* 90% interval spans one noise draw, but test–retest coverage asks it to span the difference of two independent draws (variance $2\sigma^2$); in the Gaussian reference its expected test–retest coverage is $2\Phi(z_{0.95}/\sqrt{2}) - 1 = 0.755$, *not* 0.90. This derivable gap between accuracy-coverage and repeat-coverage is precisely why the two cannot be conflated.

Distinct from the rank check. Gauge's statistic is the Spearman correlation of w with $|\Delta|$ — a monotone-ordering quantity, invariant under any positive rescaling of all widths. Our c is a scale quantity: multiplying every width by a constant leaves Spearman fixed but moves c to any value in $[0, 1]$. The two are independent axes of the same object.

4 Method self-test (SOLID)

On a synthetic test–retest harness with known truth, noise, bias, and interval scale (no upstream dependency), the statistic behaves exactly as derived (seed-fixed, n=20000): a correctly measurement-scaled 90% interval yields coverage 0.758 against the analytic 0.755; injecting a large systematic bias leaves coverage unchanged at 0.758 (the precision-not-accuracy guarantee, numerically exact); and a pure width rescale leaves the Gauge-style Spearman fixed ($0.373 \to 0.373$) while moving coverage ($0.759 \to 0.444$) — the distinctness property, demonstrated at the unit level. The statistic is therefore a working, correctly-calibrated, distinct validator before it ever touches real data.

5 Constructive counterexample: precision without coverage (SOLID)

The self-test proves the *statistic* behaves as derived; the real-data section (§6) can only *argue* that repeatability and coverage diverge, because in vivo there is no ground truth to measure coverage against. This section closes that gap with a controlled synthetic demonstration that *shows* the divergence with known truth. It is **SOLID**: it consumes only **Lattice** (synthetic ground-truth IVIM cohorts; a publication-independent resource) and **Caliper** (in-tree estimator + conformal ruler); it depends on no upstream paper.

Construction. We draw a Lattice cohort with known (D, D^*, f), acquire it *twice* from one truth (two independent Rician draws), reduce each to a whole-tumor ROI-mean over 200 voxels (Lethe's region-level posture; the $\sqrt{n_{\text{vox}}}$ precision boost), and fit both with Caliper's segmented bi-exponential estimator at single-voxel SNR 40. A split-conformal interval is calibrated on a separate bi-exponential cohort — the *believed* model — exactly Lethe's posture of calibrating on a synthetic model and deploying on reality. Because the truth is known we then measure *both* repeatability (test–retest within-subject CV and ICC of the point estimate) and *coverage* of the true parameter, marginally and conditionally on the true-D^* tercile.

The counterexample. Deploying on dispersed-perfusion reality (a velocity-dispersion truth, `stretched`, fit by the bi-exponential model — real perfusion is dispersed) produces a structural mismatch bias that ROI-averaging makes *precise* but cannot remove. The perfusion fraction f in the low-D^* regime is the clean precise-but-broken cell: it is highly repeatable (wCV $= 0.032$, ICC $= 0.995$), yet its bi-exponential- calibrated 90% interval covers the *true* f only 0.606 (BCa 95% $[0.568, 0.642]$). The matched control — the same model correctly specified (bi-exponential truth) — is equally repeatable (wCV $= 0.029$) but *not* broken (coverage 0.796). Precision is held fixed while coverage falls by 0.190; repeatability is, numerically, blind to the bias. The break is also hidden marginally — pooled f coverage looks acceptable in both arms (0.895 control / 0.840 mismatch) — and is visible only because the truth is known. This is precision $\neq$ coverage turned from asserted into shown.

Where, and why not the obvious place. The cell is f at *low* D^*, where dispersed perfusion most biases the bi-exponential fraction. Notably it is *not* the high-D^* identifiability wall: there the fit loses *precision* as well as accuracy (its repeatability fails the bar), so the wall yields an imprecise-and-biased cell, not a precise-but-biased one. A clean counterexample therefore requires a mechanism that biases *without* destroying precision — model mis-specification plus region-level averaging — and the effect grows monotonically with ROI size and SNR (better measurement *exposes*, not hides, the structural error). The matched bi-exponential and tri-exponential controls never break; the divergence is the mismatch, not IVIM *per se*.

Scope (load-bearing). This is a synthetic *possibility-and-mechanism* proof: it shows the divergence *can* occur in IVIM and *why*. It does **not** quantify the magnitude of any real-world miscalibration; the construction is a controlled demonstration, not an estimate of how mis-sized real IVIM intervals are. Read every number here as "this divergence can occur, and here is the mechanism," never as "this is how miscalibrated real IVIM is." All numbers regenerate from a fixed seed via `run_reverb.py`.

6 Real-data result (PROVISIONAL)

We fetch the genuine same-day test–retest arm of ACRIN-6698 / I-SPY2 breast DWI (TrT0/TrT1, CC-BY-4.0) on demand, reduce each repeat to a whole-tumor ROI-mean signal, and recover (D, D^*, f) per repeat; 76 tumors yield clean pairs. The conformal interval is deployed via Caliper at an effective ROI-mean signal-to-noise level (single-voxel SNR scaled by $\sqrt{n_{\text{vox}}}$), pinned before the run; the calibration regime is swept as an honesty curve. All numbers are PROVISIONAL[P] (Caliper-deployed; Fashion/Gauge/ Minos-contextualised) and regenerated by `run_validation.py`.

param	coverage [BCa 95%]	$R = \sigma_{\text{impl}}/\sigma_{\text{rep}}$	reading
D	0.263 $[0.158, 0.355]$	0.247	under-scaled (∼4× too narrow)
D^*	0.797	19.5	over-scaled (identifiability wall)
f	0.421	0.469	under-scaled

Table 1: Test–retest scale calibration on 76 ACRIN-6698 tumors at the headline single-voxel SNR $=$ 20. The analytic target for a correctly-scaled interval is 0.755.

For the well-identified D, coverage is 0.263 ($[0.158, 0.355]$) against the 0.755 target, with $R = 0.247$: the deployed interval is about four times too narrow to contain real scan–rescan variation ($\sigma_{\text{rep}} \approx 1.99 \times 10^{-4}$ vs $\sigma_{\text{impl}} \approx 0.49 \times 10^{-4}$ mm²/s; standardized residuals over- dispersed, z-disp $= 3.95$). The pre-registered sensitivity curve shows the shortfall is *robust to the calibration assumption*: coverage(D) stays well under- scaled across the entire single-voxel SNR grid (0.434 at SNR 10 down to 0.053 at SNR 100); the SNR that would reach 0.755 lies outside any plausible range. The pseudo-diffusion D^* corroborates from the other side: it is so unidentified that its interval is ∼20× *wider* than D^* repeatability (coverage 0.797, $R = 19.5$) — over-scaled by the identifiability wall. The D (under-scaled) / D^* (over-scaled-by-unidentifiability) split is itself the signature that the conformal scale is not governed by repeatability.

7 Discussion

What this validates, and what it cannot. The method correctly *measures* interval scale against repeatability, with tight CIs and a clean sensitivity curve. It validates precision only. It is, by construction, blind to accuracy: the real D estimates are themselves biased low by the high-b Rician floor, which the statistic neither sees nor corrects — and should not, because that bias cancels in Δ.

The delta versus the rank check[P]. Width *rank*-tracks repeatability ($r = 0.60$) *and* its *scale* under-covers it ($c = 0.263$, $R = 0.247$). These are consistent and complementary: an interval that orders repeatability correctly can still be the wrong size for it. This is the axis Gauge declined, and the reason this is not a duplicate of the published check.

Why under-scaled. A conformal interval is calibrated, on synthetic data, to the *modeled* (thermal) noise. Real region-level test–retest variation (within-subject CV $\sim$ 50% here) far exceeds the thermal scale because it includes repositioning, ROI re-segmentation, and noise-floor interaction — sources synthetic calibration cannot see. The honest conclusion is a *constrained validation*: repeatability does not endorse the absolute scale of conventionally-calibrated conformal IVIM intervals; it can only order, not size, them.

Limitations. The exact coverage number depends on the pinned synthetic- calibration noise model (Gaussian, thermal) and the $\sqrt{n_{\mathrm{vox}}}$ effective-SNR assumption; a Rician-floor calibration is named future work and was *not* applied post hoc (no tuning). The direction and robustness of the result — under-scaled across the whole SNR grid — do not depend on these choices. The result is region-level, not per- voxel (unregistered repeats), and inherits all Fashion/Gauge/Minos provisional caveats.

8 Conclusion

Test–retest repeatability is a legitimate but strictly bounded validator: it certifies precision, never accuracy, and it can order an interval's width without endorsing its size. Its bias-blindness is not merely argued: on synthetic ground truth (§5, SOLID) we construct a region-level estimate that is excellently repeatable yet whose interval covers the *true* parameter far below nominal, with a matched correctly-specified control that does not break — the precision $\neq$ coverage limitation, shown rather than asserted, with the caveat that it is a possibility-and-mechanism proof and not a real-world magnitude. Applied to conventionally-calibrated conformal IVIM intervals, it returns a sharply-scoped negative — the interval for D is about four times too narrow for real scan–rescan variation — which we record, honestly, as a Lethe result. On publication of Fashion, Gauge, and Minos the provisional flags are discharged by a single re-run; the promotion and fold paths are documented alongside. Notably, the retooled **Fashion** [P] reaches a parallel conclusion from the opposite direction: it now leads with an assumption-free boundary-railing diagnostic and openly scopes its calibration ruler to a ground-truth-only secondary that under-covers D^* *conditionally*, so the ruler's own paper now agrees that a ground-truth-free calibration certificate is bounded and conditional. The two bounds sit on disjoint axes — Lethe on real-data repeatability *precision*, Fashion on *synthetic* conditional coverage — and are reported as independent, not conflated.

Reproducibility & data availability. All numbers trace to seeded printouts via `consistency.py`; one command (`bash reproduce.sh`) regenerates them. Real data (ACRIN-6698, DOI `10.7937/tcia.kk02-6d95`, CC-BY-4.0) is download-on-demand; no pixel data is redistributed, only a provenance manifest. Code is MIT.